Technical Supplement to

Exact Diffusion Design for Maximally Collective Stable Turing Patterns in Binary-Complex Mass-Action Networks

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Contents

S1 Reaction matrices, flux cone, and semipositive conservation

Order species as $X_1, \ldots, X_m, Z$ and reactions as $R_0, R_2, \ldots, R_{m-2}, R_a, R_b, R_+, R_-$. For each indexed reaction, the source and target vectors have total molecularity at most two. The source matrix Y_m and stoichiometric matrix Γ_m are obtained by placing these source vectors and reaction differences in columns.

The balance equations $\Gamma_m v = 0$ read, successively, equality of every flux around the long circuit and equality of the two boundary-pair fluxes. Thus

$$\ker \Gamma_m = \operatorname{span}\{(\mathbf{1}_m, 0, 0), (0_m, 1, 1)\}.$$

The balance equations show that this kernel is exactly two-dimensional. Since Γ_m has $m+2$ columns, rank–nullity gives $\operatorname{rank} \Gamma_m = m$. As an independent explicit check, retain the rows $X_1, X_3, \ldots, X_m, Z$ and the columns $R_2, \ldots, R_{m-2}, R_a, R_b, R_+$, with the chain range empty when $m = 3$. The resulting maximal minor has determinant $4(-1)^m$. The vector

$$c = (0, 4, \ldots, 4, 2, 1)^T$$

is orthogonal to every reaction vector, so $\operatorname{im} \Gamma_m = c^\perp$. Notice that c is semipositive rather than positive: X_1 is absent.

For flux $(a\mathbf{1}_m, b, b)$ and unit equilibrium, $A_m(a,b) = \Gamma_m \operatorname{diag}(a\mathbf{1}_m, b, b) Y_m^T$ has precisely the entries printed in the main text. At an arbitrary positive equilibrium, right multiplication by $H = \operatorname{diag}(1/x_i^*)$ gives $J = A_m(a,b)H$. Conversely the rate assignment $k_r = v_r/(x^*)^{y_r}$ realizes every $a, b, H > 0$.

S2 All-spectrum principal-block classification

Use the directed convention $X_j \to X_i$ when the (i,j) Jacobian entry is nonzero. The interior arrows are the one-way chain $X_2 \to X_3 \to \cdots \to X_{m-1}$. The boundary arrows are those incident to X_1, X_m, Z and the returns $X_{m-1} \to X_1$, $X_m \to X_2$. The following case analysis is exhaustive for an induced set with fewer than m vertices.

For $m = 3$, inspect the principal sets of order at most two directly. The sets $\{X_1, X_2\}$ and $\{X_2, X_3\}$ are the two complete long cycles; every other nontrivial strongly connected component is a principal subgraph of $\{X_1, X_3, Z\}$. Hence assume $m \geq 4$ in the chain argument below.

1. If a component contains a chain arrow and returns through $X_{m-1} \to X_1$, then the unique forward path from X_1 forces every vertex $X_1, \dots, X_{m-1}$. This is the first complete long cycle.
2. If it contains a chain arrow and returns through $X_m \to X_2$, then the forward path forces every vertex $X_2, \dots, X_m$. This is the second complete long cycle.
3. The alternative chain closure $X_1 \to X_2 \to \cdots \to X_m \to X_1$ uses m vertices and is excluded. Adding any boundary vertex to either complete long cycle also uses at least m vertices.
4. If neither long cycle is complete, every feedback path not involving a complete chain lies inside $\{X_1, X_m, Z\}$. All other retained chain vertices have no return path and therefore form negative singleton diagonal blocks.

At $b = 2a$, the coefficient A_{m1} vanishes. This deletes the arrow $X_1 \to X_m$; edge deletion can split but cannot merge strongly connected components, so the same list remains exhaustive.

For the boundary triad, the Routh–Hurwitz difference expands as

$$\begin{aligned} c_1c_2 - c_3 = a\big(&4a^2h_1^2h_m + 16a^2h_1h_m^2 + 11abh_1^2h_m + 4abh_1^2h_Z \\ &+ 32abh_1h_m^2 + 32abh_1h_Zh_m + 64abh_Zh_m^2 + 7b^2h_1^2h_m \\ &+ 4b^2h_1^2h_Z + 7b^2h_1h_m^2 + 48b^2h_1h_Zh_m + 16b^2h_1h_Z^2 \\ &+ 16b^2h_Zh_m^2 + 64b^2h_Z^2h_m\big). \end{aligned}$$

Every monomial is strictly positive. The two long-cycle modulus inequalities are strict because $a + b > a$ and $4a + b > 4a$. Frobenius permutation therefore proves Hurwitz stability of every smaller principal block.

Sparse expansion on $\{X_1, \dots, X_m\}$ gives

$$(-1)^m \det J_{\widehat{Z},\widehat{Z}} = -2a^{m-1}b\prod_{i=1}^{m} h_i.$$

The complete omission table is derived in Section S4 below.

S3 The principal-minor diffusion-ray theorem in full detail

Let $n \geq 2$, $a_I = (-1)^{|I|}\det J_{I,I}$, and $a_\varnothing = 1$. Assume $\det J = 0$, $a_I > 0$ for $|I| \leq n-2$, and $\sum_{|I|=n-1} a_I > 0$. The last condition follows from a simple zero and stable complementary spectrum but is stated directly because it is all the proof uses. Multilinearity in the diagonal entries of sD gives

$$\det(sD - J) = \sum_{I\subseteq[n]} a_I \prod_{j\notin I} sd_j.$$

Regrouping by $k = n - |I|$ yields

$$\det(sD - J) = \sum_{k=1}^{n} \beta_k(D)s^k, \qquad \beta_k(D) = \sum_{|I|=n-k} a_I \prod_{j\notin I} d_j.$$

For $k \geq 2$, every contributing set has order at most $n-2$, so the stated coefficient hypothesis gives $a_I > 0$ and hence $\beta_k > 0$.

The bracket $q_D(s) = \beta_1 + \beta_2 s + \cdots + \beta_n s^{n-1}$ is strictly increasing for $s > 0$. If $\beta_1 < 0$, it has exactly one positive zero and its derivative there is positive. This proves scalar simplicity of the dispersion root. Ordinary algebraic simplicity of zero as an eigenvalue of $J - s_*D$ follows from the characteristic-polynomial derivative below.

For the positive-real-eigenvalue band, differentiate

$$\chi_s(\lambda) = \sum_I a_I \prod_{j\notin I}(\lambda + sd_j).$$

Every order-at-most-$(n-2)$ contribution is nonnegative for $\lambda \geq 0$. The order-$(n-1)$ contribution to χ_s' is the constant $\sum_{|I|=n-1} a_I$. This is positive by hypothesis. Under the common spectral corollary it equals the coefficient of s in $\det(sI - J)$, and equal damping sends the simple zero and all stable eigenvalues left. Thus $\chi_s' > 0$ on the nonnegative real axis. The sign of $\chi_s(0)$ then determines whether there is exactly one positive real eigenvalue. At $s = s_*$, $\chi_{s_*}(0) = 0$ and $\chi_{s_*}'(0) > 0$, so zero is an algebraically simple eigenvalue of $J - s_*D$.

The theorem says nothing about nonreal unstable eigenvalues when $s > s_*$. The all-mode certificates below are separate, profile-specific results.

S4 Order-$(n-1)$ omission minors and diffusion design

Right multiplication by H multiplies a principal determinant by the product of its retained h_i, so the combinatorial coefficients may be computed at $H = I$.

Omitting Z leaves the complete X-chain block. Its sparse cycle-cover expansion gives

$$(-1)^m \det J_{\widehat{Z},\widehat{Z}} = -2a^{m-1}b \prod_{i=1}^{m} h_i.$$

For $2 \le j \le m-1$, omit X_j and order the remaining vertices by the two feed-forward chain fragments, followed by $\{X_1, X_m, Z\}$. The resulting matrix is block triangular. The $m-3$ chain singletons contribute signed factors ah_i, and the boundary triad contributes

$$(-1)^3 \det J_{\{X_1,X_m,Z\},\{X_1,X_m,Z\}} = 16a^2 b\, h_1 h_m h_Z.$$

Therefore

$$(-1)^m \det J_{\widehat{X_j},\widehat{X_j}} = 16a^{m-1} b h_Z \prod_{\substack{1\le i\le m\\ i\ne j}} h_i.$$

Omitting X_1 leaves the restricted semipositive conservation vector in the left kernel. Omitting X_m leaves the restriction of $H^{-1}(2,-2,\ldots,-2,0,1)^T$ in the right kernel, because the omitted coordinate is zero. Hence both determinants vanish.

Multiplication by the omitted diffusivities produces

$$\beta_1(D) = 2a^{m-1}b \left(\prod_{i=1}^{m} h_i\right)\left(8h_Z \sum_{j=2}^{m-1} \frac{d_j}{h_j} - d_Z\right).$$

This yields the exact diffusion criterion and explains the coefficient eight as the exact ratio 16/2.

Define the stable realization domain

$$\mathfrak{S}_m = \{(a,b,H) : a,b>0,\ H \succ 0,\ A_m(a,b)H|_{c^\perp} \text{ is Hurwitz}\}.$$

The complete region $\mathfrak{S}_m$ is not characterized here. For any $(a,b,H)\in\mathfrak{S}_m$, if $d_{\min} = \min_i d_i$, then

$$d_Z > 8h_Z \sum_{j=2}^{m-1} \frac{d_j}{h_j} \ge d_{\min} T(H),$$

so $\chi_D > T(H)$. Setting all X diffusivities to one and $d_Z = T(H)+\varepsilon$ proves the nonattained fixed-H infimum. Since $h_Z/h_j \ge 1/\chi_H$, one obtains $\chi_D\chi_H > 8(m-2)$. The implication $T(H) > 1$ is necessary under homogeneous stability; its converse is not proved.

S5 Improved unit-equilibrium critical profile

Put $K_i = 91m - 181 - i$. The right vector $r = (r_1,\ldots,r_m,r_Z)^T$ and diffusion profile are

$$r_1 = 1,\quad r_i = -\frac{K_i}{63(m-2)},\quad r_m = -\frac{2}{9},\quad r_Z = \frac{5}{14},$$

$$d_1 = \frac{23}{63},\quad d_i = \frac{1}{K_i},\quad d_m = \frac{1}{7},\quad d_Z = \frac{16}{45}.$$

Direct substitution gives $(A_m - D_m)r = 0$. With $\ell_Z = 1$, the left vector $\ell = (\ell_1,\ldots,\ell_m,\ell_Z)^T$ is

$$\ell_1 = -\frac{266}{815},\quad \ell_i = \frac{78260(m-2)}{163(91m-180-i)},\quad \ell_m = \frac{18368}{7335},\quad \ell_Z = 1.$$

Define

$$\mathfrak{h}_m = \sum_{j=1}^{m-2} \frac{1}{91m-181-j}.$$

Then

$$\ell^T r = -\frac{7043400m - 13600927 - 7043400\mathfrak{h}_m}{924210} < 0,$$

$$\ell^T D_m r = -\frac{2(1760850\mathfrak{h}_m + 16559)}{462105} < 0.$$

The harmonic bound $\mathfrak{h}_m \le (m-2)/(90m-179)$ makes the first sign an elementary shifted-polynomial inequality.

Homogeneous and spatial determinants

The homogeneous determinant has

$$\det(\lambda I - A_m) = \lambda q_m(\lambda), \qquad \lambda q_m = (1+\lambda)^{m-3}P(\lambda) - R(\lambda),$$

with $P = \lambda^4 + 12\lambda^3 + 42\lambda^2 + 47\lambda + 16$ and $R = 5\lambda^2 + 33\lambda + 16$. For $m = 3$,

$$q_3 = (\lambda+7)(\lambda^2 + 5\lambda + 2)$$

is Hurwitz. For $m \ge 4$, the exact 35-term certificate for $|1+\lambda|^2|P|^2 - |R|^2$ is strictly positive away from the origin in the closed right half-plane. Since $|1+\lambda| \ge 1$ there, a nonzero closed-right-half-plane root of q_m would contradict $(1+\lambda)^{m-3}P = R$. Finally, $q_m(0) = 16m - 34 > 0$, so the conservation zero is simple and the complementary homogeneous spectrum is stable for every $m \ge 3$. For a mode with damping factor t,

$$\det(\lambda I - A_m + tD_m) = Q_m F - G,$$

where

$$Q_m = \prod_{i=2}^{m-1}\left(\lambda + 1 + \frac{t}{K_i}\right),$$

$$g_1 = \lambda + 2 + \frac{23}{63}t, \quad g_m = \lambda + 5 + \frac{1}{7}t, \quad g_Z = \lambda + 4 + \frac{16}{45}t,$$

$$F = g_1 g_m g_Z - 4g_1 - 4g_m + g_Z, \qquad G = g_Z(4g_1 + g_m) - 36.$$

At onset, $\prod_{i=2}^{m-1}(1+K_i^{-1}) = 91/90$. The exact certificate tables appear in Section S8. The equality point is algebraically simple, not merely isolated from nonzero closed-right-half-plane roots. Indeed, with

$$\Pi_m(\lambda) = \det(\lambda I - A_m + D_m) = Q_m(\lambda,1)F(\lambda,1) - G(\lambda,1),$$

one has the all-dimensional identity

$$\Pi_m'(0) = \frac{7043400m - 13600927 - 7043400\mathfrak{h}_m}{255150} = -\frac{163}{45}\,\ell^T r > 0.$$

Thus zero is algebraically simple and the kernel and cokernel of $A_m - D_m$ are one-dimensional.

S6 Conservation-compatible stable-mode corrections

The quadratic tensor is obtained reaction by reaction. Its components are

$$\begin{aligned}
(B(u,v))_1 &= -(u_1 v_{m-1} + u_{m-1}v_1) + 2u_Z v_Z - (u_1 v_m + u_m v_1),\\
(B(u,v))_2 &= -(u_1 v_2 + u_2 v_1) + 2u_m v_m,\\
(B(u,v))_i &= u_1 v_{i-1} + u_{i-1}v_1 - u_1 v_i - u_i v_1,\\
(B(u,v))_m &= 2(u_1 v_{m-1} + u_{m-1}v_1) - 4u_m v_m + 2u_Z v_Z - (u_1 v_m + u_m v_1),\\
(B(u,v))_Z &= -4u_Z v_Z + 2(u_1 v_m + u_m v_1).
\end{aligned}$$

Here the third line holds for $3 \le i \le m-1$. Reaction-wise conservation implies $c^T B(u,v) = 0$.

The zero mode solves

$$A_m w_0 = -\frac{1}{4}B(r,r), \qquad c^T w_0 = 0.$$

The exact solution is

$$(w_0)_1 = \frac{182448m - 373417}{31752(8m-17)},$$

$$(w_0)_2 = \frac{1008m^2 - 20459m + 37138}{31752(m-2)(8m-17)},$$

$$(w_0)_i = (w_0)_2 - \frac{i-2}{126(m-2)}, \quad (w_0)_m = -\frac{1}{81}, \quad (w_0)_Z = \frac{16861m - 34044}{7938(8m-17)}.$$

The second harmonic solves

$$(A_m - 4D_m)w_2 = -\frac{1}{4}B(r,r).$$

With

$$\sigma = \frac{1}{126(m-2)}, \qquad T_i = \frac{K_{i-3}K_{i-2}K_{i-1}K_i}{K_{-1}K_0K_1K_2},$$

the chain recurrence is

$$(w_2)_i = T_i\left((w_2)_2 + \frac{\sigma K_2}{3}\right) - \frac{\sigma K_i}{3}.$$

Its boundary denominator is

$$\mathcal{Q}_m = 589180301m^3 - 3500015940m^2 + 6930529579m - 4574434500.$$

After $m = u+3$, the coefficients are

$$589180301, \quad 1802606769, \quad 1838302066, \quad 624878904,$$

so it never vanishes for $m \geq 3$. The complete boundary solve is exposed as follows. Write

$$W_1 = (w_2)_1, \quad W_2 = (w_2)_2, \quad W_m = (w_2)_m, \quad W_Z = (w_2)_Z, \qquad b^{(2)} = -\frac{1}{4}B(r,r),$$

and set

$$\delta_m = \frac{\sigma}{3}\left(T_{m-1}K_2 - K_{m-1}\right).$$

Substitution of the recurrence into the four boundary rows gives the complete system

$$\begin{pmatrix} -218/63 & -T_{m-1} & -1 & 2 \\ -1 & -(1+4/K_2) & 2 & 0 \\ 1 & 2T_{m-1} & -39/7 & 2 \\ 2 & 0 & 2 & -244/45 \end{pmatrix} \begin{pmatrix} W_1 \\ W_2 \\ W_m \\ W_Z \end{pmatrix} = \begin{pmatrix} b_1^{(2)} + \delta_m \\ b_2^{(2)} \\ b_m^{(2)} - 2\delta_m \\ b_Z^{(2)} \end{pmatrix}.$$

Its determinant is exactly

$$\frac{64\mathcal{Q}_m}{6615(91m-183)(91m-181)(91m-180)}.$$

All factors in the denominator are positive for $m \geq 3$, while the displayed shifted-positive certificate proves $\mathcal{Q}_m > 0$. Thus the four boundary values are the unique solution of this displayed system. Their expanded rational forms are supplied in the machine-readable archive for convenient verification.

The cubic coefficient is

$$c_m = \frac{\ell^T\{B(r,w_0) + \frac{1}{2}B(r,w_2)\}}{\ell^T r}.$$

This is the cubic coefficient of the one-dimensional center-manifold normal form in the adjoint-projection coordinate. Setting the reduced vector field to zero gives the associated stationary Lyapunov–Schmidt equation. The spatial reflection $(\mathcal{R}u)(\xi) = u(\pi - \xi)$ commutes with the Neumann semiflow, preserves the fixed integrated-mass class, and sends $r\cos\xi$ to $-r\cos\xi$. On an equivariant center manifold the adjoint-projection action is $\mathcal{R} : A \mapsto -A$; hence the reduced vector field is odd in A, and reflection exchanges the two nonzero branches. For the cubic numerator, introduce

$$P_R(m) = 686050404808142087 68m^4 - 550882186169626030957m^3$$
$$+ 165861263293744967085 2m^2 - 221922647620410350132 3m$$

$$+ 111337927497580956570 0,$$

and

$$\begin{aligned} P_C(m) = 652054120726848m^4 - 5151971981328467m^3 \\ + 15265080924982572m^2 - 20102347725659113m \\ + 9927281930180400. \end{aligned}$$

Then the exact decomposition is

$$N_m = N_m^{\mathrm{ref}} = R_m + C_m \mathfrak{h}_m,$$

where

$$R_m = \frac{P_R(m)}{286118780220(8m-17)\mathcal{Q}_m}, \qquad C_m = -\frac{215 P_C(m)}{11645046(8m-17)\mathcal{Q}_m}.$$

The shifted-positive coefficient certificates in Section S8.2 show that $\mathcal{Q}_m > 0$, $P_R(m) > 0$, and $P_C(m) > 0$ for $m \geq 3$. Consequently $R_m > 0$ and $C_m < 0$; in the latter conclusion the external factor -215 reverses the sign of the positive polynomial P_C.

The exact clearing identity is

$$R_m + C_m \frac{m-2}{90m-179} = \frac{L_m}{286118780220(8m-17)(90m-179)\mathcal{Q}_m},$$

where

$$\begin{aligned} L_m = 272994514782766788672 0m^5 - 27755132420474170999952m^4 \\ + 112813395868533457497683m^3 - 229153280695458887386228m^2 \\ + 232620996871721820873517m - 94412163900120968220300. \end{aligned} \tag{S6.1}$$

After $m = u + 3$, its coefficients are

$$\begin{gathered} 2729945147827667886720, 13194044796940847300848, 25446870127333515303059, \\ 24475321329207325509911, 11736484608366875120374, 2243933770033305838488, \end{gathered}$$

all positive. Since $C_m < 0$ and $\mathfrak{h}_m \leq (m-2)/(90m-179)$, the complete comparison is

$$R_m + C_m \mathfrak{h}_m \geq R_m + C_m \frac{m-2}{90m-179} > 0.$$

Thus the cubic numerator is positive and, since $\ell^T r < 0$, $c_m < 0$.

S7 Equilibrium-scaled stable family

Let $\nu = m - 2$ and define

$$\kappa_\nu = \begin{cases} 1/\sqrt{3}, & \nu = 1, \\ \sqrt{5}/2, & \nu \geq 2, \end{cases} \qquad L_0(\nu) = \frac{\kappa_\nu}{\sqrt{\nu}}, \qquad L_1(\nu) = \frac{90\nu}{90\nu + 1}.$$

This interval is nonempty. For $\nu = 1$, square both positive sides and use $8281 < 24300$. For $\nu \geq 2$,

$$(90\nu + 1)^2 \leq (91\nu)^2 < 6480\nu^3,$$

which is equivalent to $L_0(\nu) < L_1(\nu)$. The lower endpoints are sufficient boundaries of the certificates below, not proved intrinsic dynamical boundaries. For $L \in [L_0, L_1]$, set

$$h_1 = h_m = h_Z = 1, \qquad h_i = \frac{K_i}{L K_{i-1}}.$$

Write $\mathsf{H}_m(L) = \mathrm{diag}(h_1, \ldots, h_m, h_Z)$. The physical equilibrium is $x^* = \mathsf{H}_m(L)^{-1}\mathbf{1}$ and the physical diffusion matrix is $D^{\mathrm{phys}} = \mathsf{H}_m(L)\Delta$, where Δ is the improved unit profile. In normalized variables $\widehat{x} = \mathsf{H}_m(L)x$,

$$\partial_t \widehat{x} = \mathsf{H}_m(L)\{f_m(\widehat{x}) + (1-\mu)\Delta \partial_{\xi\xi}\widehat{x}\}.$$

At $\mu = 0$, the stationary bracket $f_m(\widehat{x}) + \Delta\partial_{\xi\xi}\widehat{x}$ and its critical right vector are unchanged. The dynamic left vector is

$$\widetilde{\ell}(L) = \mathsf{H}_m(L)^{-1}\ell,$$

and the physical conservation gauge uses $\mathsf{H}_m(L)^{-1}c$. The crossing numerator is unchanged by the row scaling:

$$\widetilde{\ell}(L)^T\mathsf{H}_m(L)\Delta r = \ell^T\Delta r < 0.$$

Together with $\widetilde{\ell}(L)^T r < 0$, this gives

$$\eta_m(L) = \frac{\widetilde{\ell}(L)^T\mathsf{H}_m(L)\Delta r}{\widetilde{\ell}(L)^T r} > 0.$$

Moreover,

$$\ker[\mathsf{H}_m(L)(A_m - \Delta)] = \ker(A_m - \Delta) = \operatorname{span}\{r\},$$

because $\mathsf{H}_m(L)$ is invertible and the unit critical zero is simple. If $\mathsf{H}_m(L)(A_m - \Delta)v = r$ for a generalized eigenvector v, then left multiplication by $\widetilde{\ell}(L)^T$ would imply $0 = \widetilde{\ell}(L)^T r$, a contradiction. Hence the scaled critical zero is algebraically simple as well.

For interior chain factors,

$$g_i = \frac{K_{i-1}}{K_i}(1 + L\lambda) + \frac{t-1}{K_i}.$$

When $t \geq 1$ and $\Re\lambda \geq 0$,

$$|Q_m|^2 \geq \left(\frac{91}{90}\right)^2 |1 + L\lambda|^{2\nu} \geq \left(\frac{91}{90}\right)^2 \left(1 + 2\nu L\Re\lambda + \frac{1}{3}(\Im\lambda)^2\right).$$

The last inequality holds because $\nu L^2 \geq 1/3$ throughout the certified interval. The 84-term spatial coefficient table in Section S8 is nonnegative and is strict except at $(\lambda, t) = (0, 1)$.

Homogeneous stability uses a different boundary polynomial. At $t = 0$,

$$\frac{\det(\lambda I - \mathsf{H}_m(L)A_m)}{\det \mathsf{H}_m(L)} = Q(\lambda)F_0(\lambda) - R(\lambda),$$

where

$$Q = \prod_{i=2}^{m-1}\left(1 + L\frac{K_{i-1}}{K_i}\lambda\right), \qquad F_0 = \lambda^3 + 11\lambda^2 + 31\lambda + 16, \qquad R = 5\lambda^2 + 33\lambda + 16.$$

For $\Re\lambda = x \geq 0$, every factor in Q has modulus at least $|1 + L\lambda|$. If $\nu \geq 2$, then

$$|Q|^2 \geq |1 + L\lambda|^{2\nu} \geq 1 + 2\nu Lx + \nu L^2(\Im\lambda)^2 \geq 1 + Ax + \frac{5}{4}(\Im\lambda)^2, \qquad A = 2\nu L.$$

Here $A \geq \sqrt{5\nu} \geq \sqrt{10} > 1/4$. The exact polynomial

$$\left(1 + Ax + \frac{5}{4}(\Im\lambda)^2\right)|F_0|^2 - |R|^2$$

has 22 nonzero monomials. With $U = A - 1/4$, its coefficient polynomials have nonnegative rational coefficients and the expression is strict away from $\lambda = 0$. Thus there is no nonzero homogeneous eigenvalue in the closed right half-plane. The coefficient 5/4 is sharp for this coefficientwise certificate. If it is replaced by B, then at $x = 0$ the coefficient of $z = y^2$ in $(1 + Bz)|F_0|^2 - |R|^2$ is $256B - 320$. It vanishes at $B = 5/4$ and is negative for $B < 5/4$. This does not show that L_0 is an intrinsic dynamical boundary; it identifies only the boundary of the present exact modulus certificate. Moreover

$$(QF_0 - R)'(0) = 16L\sum_{i=2}^{m-1}\frac{K_{i-1}}{K_i} - 2 > 0,$$

where the inequality uses $K_{i-1}/K_i > 1$ and $L \geq \sqrt{5}/(2\sqrt{\nu})$. Thus the conservation zero is simple. If $\nu = 1$, put $\gamma = 91L/90$. Direct division by the conservation factor gives

$$\frac{QF_0 - R}{\lambda} = \gamma\lambda^3 + (1 + 11\gamma)\lambda^2 + (6 + 31\gamma)\lambda + (16\gamma - 2).$$

Here $\gamma > 1/8$, and the remaining cubic Routh–Hurwitz difference is $6 + 99\gamma + 325\gamma^2 > 0$. This proves homogeneous relative stability in the exceptional dimension as well.

Let $\rho_m = (2, -2, \dots, -2, 0, 1)^T$. The physical zero-mode gauge is

$$w_0(L) = w_0^{\mathrm{ref}} + \tau_m(L)\rho_m, \qquad \tau_m(L) = -\frac{(\mathsf{H}_m(L)^{-1}c)^T w_0^{\mathrm{ref}}}{(\mathsf{H}_m(L)^{-1}c)^T \rho_m}.$$

The second harmonic is unchanged. The cubic numerator satisfies

$$N_m(L) = N_m^{\mathrm{ref}} + \tau_m(L) S_m,$$

where $N_m^{\mathrm{ref}} = R_m + C_m \mathfrak{h}_m$. To make its quantitative margin explicit, set

$$\begin{aligned} P_{\mathrm{ref}}(\nu) = {} & 379050298663726568484 0\nu^5 - 974216530468600286489\nu^4 \\ & - 531035674409212188 71\nu^3 - 576386186827093561\nu^2 \\ & + 3649732858601219\nu + 55281268032918 \end{aligned}$$

and

$$\begin{aligned} D_{\mathrm{ref}}(\nu) = {} & 715296950550(8\nu - 1)(90\nu + 1) \\ & \times (589180301\nu^3 + 35065866\nu^2 + 629431\nu + 3306). \end{aligned}$$

Since $C_m < 0$ and $\mathfrak{h}_m \le \nu/(90\nu + 1)$, exact clearing gives

$$N_m^{\mathrm{ref}} - \frac{1}{100} \ge \frac{P_{\mathrm{ref}}(\nu)}{D_{\mathrm{ref}}(\nu)}.$$

The denominator is positive for $\nu \ge 1$. After $\nu = v + 1$, the coefficients of P_{ref}, in descending powers $v^5, \dots, v^0$, are

$$\begin{gathered} 3790502986637265684840, 1797829840271772813771 1, 3395506017705733448357 3, \\ 3189984359505146437929 2, 1489518898634836803572 8, 276261020755504372005 6. \end{gathered}$$

They are positive, so $P_{\mathrm{ref}}(\nu) > 0$ and $N_m^{\mathrm{ref}} > 1/100$.

Separately,

$$S_m = -\frac{4(1760850\mathfrak{h}_m - 10253)}{462105}.$$

The bounds $1/91 \le \mathfrak{h}_m < 1/90$ give $S_m < 0$ because

$$\frac{1760850}{91} - 10253 > 0,$$

and give $S_m > -1/10$ because

$$\frac{4}{462105}\left(\frac{1760850}{90} - 10253\right) < \frac{1}{10}.$$

For the remaining gauge estimate, define

$$\begin{aligned} A_\tau = {} & 1494249120\mathfrak{h}_m L\nu^2 - 69786990\mathfrak{h}_m L\nu + 108738630L\nu^2 \\ & + 1214388L\nu - 8521L - 125249670\nu^2 + 1031940\nu, \end{aligned}$$

$$B_\tau = 32760\mathfrak{h}_m L\nu + 32760L\nu^2 + 4L - 4095\nu.$$

Then the closed form is

$$\tau_m(L) = -\frac{A_\tau}{15876(8\nu - 1)B_\tau}.$$

On the broader certified cubic range $L \ge 1/\sqrt{3\nu}$,

$$B_\tau = 4095\nu\big[8L(\nu + \mathfrak{h}_m) - 1\big] + 4L > 0,$$

because $L\nu \geq 1/\sqrt{3}$. Exact differentiation then gives

$$\partial_{\mathfrak{h}}\tau_m = -\frac{4225L\nu^2(182448L\nu + 1008L - 7513)}{2B_\tau^2} < 0,$$

$$\partial_L\tau_m = -\frac{65\nu(-61531470\mathfrak{h}_m\nu + 125249670\nu^2 + 1031940\nu - 7513)}{252B_\tau^2} < 0.$$

For the first sign, $L\nu \geq 1/\sqrt{3} > 1/2$ makes the parenthesis positive. For the second, use $\mathfrak{h}_m < 1/90$ and $\nu \geq 1$. Thus the maximum occurs at $\mathfrak{h}_m = 1/91$ and $L = 1/\sqrt{3\nu}$.

Put $y = \sqrt{3\nu}$ and define the unreduced endpoint polynomial

$$\begin{aligned} P_\tau(\nu, y) = & -1040195520\nu^3 + 756272790\nu^2 y - 507201030\nu^2 \\ & -21412755\nu y - 935658\nu + 58481 \end{aligned}$$

and $D_\tau(\nu, y) = -32760\nu^2 + 4095\nu y - 360\nu - 4$. Direct substitution gives

$$\tau_m(1/91, 1/y) - \frac{1}{20} = -\frac{P_\tau(\nu, y)}{79380(8\nu - 1)D_\tau(\nu, y)}.$$

Since $y < 2\nu$, $D_\tau < 0$. For $\nu = 1$, the coefficient of y in $P_\tau(1, y)$ is positive and $\sqrt{3} < 7/4$, whence

$$P_\tau(1, \sqrt{3}) < P_\tau(1, 7/4) = -\frac{1049074663}{4} < 0.$$

For $\nu \geq 2$, $y \leq 5\nu/4$ and deletion of the negative term $-21412755\nu y$ gives $P_\tau(\nu, y) < P_{\mathrm{up}}(\nu)$, where

$$P_{\mathrm{up}}(\nu) = -\frac{189709065}{2}\nu^3 - 507201030\nu^2 - 935658\nu + 58481.$$

This upper-bound polynomial is human-checkable:

$$P'_{\mathrm{up}}(\nu) = -\frac{569127195}{2}\nu^2 - 1014402060\nu - 935658 < 0 \qquad (\nu \geq 0),$$

and

$$P_{\mathrm{up}}(2) = -2789453215 < 0.$$

Hence $P_{\mathrm{up}}(\nu) < 0$ for every $\nu \geq 2$, and therefore $P_\tau(\nu, y) < 0$ in all dimensions. Because the displayed denominator $79380(8\nu - 1)D_\tau$ is negative, the exact endpoint comparison proves $\tau_m(L) < 1/20$.

If $\tau_m(L) \leq 0$, then $\tau_m(L)S_m \geq 0$. If instead $0 < \tau_m(L) < 1/20$, then $\tau_m(L)S_m > -1/200$. Together with $N_m^{\mathrm{ref}} > 1/100$, this yields $N_m(L) > 1/200$. Also

$$\ell^T \mathsf{H}_m(L)^{-1} r = -\frac{485873}{924210} - \frac{11180}{1467}L(m-2) < 0.$$

Thus the scaled cubic coefficient is explicitly

$$c_m(L) = \frac{N_m(L)}{\tilde{\ell}(L)^T r} = \frac{N_m(L)}{\ell^T \mathsf{H}_m(L)^{-1} r} < 0.$$

Therefore the family is uniformly supercritical over the certified interval. Indeed, the cubic comparison was proved on the broader range $L \geq 1/\sqrt{3\nu}$; the present lower endpoint agrees with this value for $\nu = 1$ and is larger for $\nu \geq 2$. The diagonal concentration scaling commutes with the same spatial reflection $\xi \mapsto \pi - \xi$. Hence the scaled center-manifold vector field remains odd in its adjoint-projection amplitude, and reflection exchanges the two nonzero patterned branches.

The contrasts are

$$\chi_D(L) = \frac{23}{63}(91\nu L), \qquad \chi_H(L) = \frac{91\nu - 1}{91\nu L},$$

where χ_H measures equilibrium concentration-scale separation, not a universal measure of biological expense or implausibility. Their product is fixed:

$$\chi_D(L)\chi_H(L) = \frac{23(91\nu - 1)}{63} = \frac{23(91m - 183)}{63} = \chi_D^{\mathrm{unit}}(m).$$

Thus L reallocates contrast between diffusion and equilibrium concentration scales and reduces $\max(\chi_D, \chi_H)$, not their product. At $L = L_0$,

$$\chi_D = \frac{2093}{63}\kappa_\nu\sqrt{\nu}, \qquad \chi_H = \frac{\sqrt{\nu}}{\kappa_\nu}\frac{91\nu - 1}{91\nu},$$

so both contrasts have square-root growth.

S8 Exact coefficient tables

S8.1 Coefficientwise-nonnegative modulus certificates

The first four tables below are coefficientwise-nonnegative modulus certificates. Each row records an exponent vector and the exact coefficient of the corresponding polynomial. The later shifted-polynomial tables are signed scalar certificates; for each such table we state the polynomial, any overall sign prefactor, the parameter domain, and the sign conclusion separately.

By conjugation, take $y \geq 0$ and use the variables

$$\lambda = x + iy = x + i\sqrt{z}, \qquad z = y^2, \qquad t = 1 + s, \qquad x, y, z, s \geq 0.$$

For the unit-profile spatial calculation, define

$$g_1 = \lambda + 2 + \frac{23}{63}t, \qquad g_m = \lambda + 5 + \frac{1}{7}t, \qquad g_Z = \lambda + 4 + \frac{16}{45}t,$$

$$F = g_1 g_m g_Z - 4g_1 - 4g_m + g_Z, \qquad G = g_Z(4g_1 + g_m) - 36.$$

The 77-term table is exactly the expansion in (x, z, s) of

$$E_{77}(x, z, s) = \left[\left(\frac{91}{90}\right)^2 + z\right] \left|F(x + i\sqrt{z}, 1 + s)\right|^2 - \left|G(x + i\sqrt{z}, 1 + s)\right|^2.$$

Every one of its 77 displayed coefficients is strictly positive; there is no constant term. Hence equality occurs only at $x = z = s = 0$, equivalently $(\lambda, t) = (0, 1)$.

For the equilibrium-scaled spatial calculation, put $A = 2\nu L > 0$. The 84-term table is exactly the expansion in (x, z, s) of

$$E_{84}(x, z, s; A) = \left(\frac{91}{90}\right)^2 \left(1 + Ax + \frac{z}{3}\right) \left|F(x + i\sqrt{z}, 1 + s)\right|^2 - \left|G(x + i\sqrt{z}, 1 + s)\right|^2.$$

Its 84 coefficients are polynomials in A with nonnegative rational coefficients and are positive for $A > 0$. Again there is no constant term, and equality occurs only at $x = z = s = 0$, equivalently $(\lambda, t) = (0, 1)$.

For the equilibrium-scaled homogeneous calculation, define

$$F_0(\lambda) = \lambda^3 + 11\lambda^2 + 31\lambda + 16, \qquad R(\lambda) = 5\lambda^2 + 33\lambda + 16, \qquad U = A - \frac{1}{4}.$$

The 22-term table is exactly the expansion in (x, z) of

$$E_{22}(x, z; U) = \left[1 + \left(U + \frac{1}{4}\right)x + \frac{5}{4}z\right] |F_0(x + i\sqrt{z})|^2 - |R(x + i\sqrt{z})|^2.$$

For $U \geq 0$, all 22 coefficient polynomials are nonnegative, and the expression vanishes only at $x = z = 0$, equivalently $\lambda = 0$.

Finally, with

$$P(\lambda) = \lambda^4 + 12\lambda^3 + 42\lambda^2 + 47\lambda + 16,$$

the unit-profile homogeneous table is the 35-term expansion in (x, z) of

$$E_{35}(x, z) = |1 + \lambda|^2 |P(\lambda)|^2 - |R(\lambda)|^2, \qquad \lambda = x + i\sqrt{z}.$$

All 35 coefficients are strictly positive, there is no constant term, and equality occurs only at $x = z = 0$, equivalently $\lambda = 0$.

From the repository root, the command `python independent_verifier/verify_symbolic_certificates.py` regenerates and checks the tables from exact symbolic expressions.

35-term homogeneous certificate

$\deg_x$	$\deg_z$	coefficient
10	0	1
9	0	26
8	1	5
8	0	277
7	1	104
7	0	1582
6	2	10
6	1	892
6	0	5356
5	2	156
5	1	4034
5	0	11282
4	3	10
4	2	1014
4	1	10432
4	0	15116
3	3	104
3	2	3322
3	1	15628
3	0	12612
2	4	5
2	3	460
2	2	5804
2	1	13296
2	0	5568
1	4	26
1	3	870
1	2	4858
1	1	5700
1	0	960
0	5	1
0	4	61
0	3	728
0	2	1508
0	1	192

77-term improved-profile spatial certificate

$\deg_x$	$\deg_z$	$\deg_s$	coefficient
6	1	0	1
6	0	0	$\frac{8281}{8100}$
5	1	1	$\frac{544}{315}$
5	1	0	$\frac{7474}{315}$
5	0	1	$\frac{160888}{91125}$
5	0	0	$\frac{4420871}{182250}$
4	2	0	3
4	1	2	$\frac{40058}{33075}$
4	1	1	$\frac{1147126}{33075}$
4	1	0	$\frac{9681647}{44100}$
4	0	2	$\frac{3384901}{2733750}$
4	0	1	$\frac{96932147}{2733750}$
4	0	0	$\frac{1073317517}{5467500}$
3	2	1	$\frac{1088}{315}$
3	2	0	$\frac{14948}{315}$
3	1	3	$\frac{2744576}{6251175}$

$\deg_x$	$\deg_z$	$\deg_s$	coefficient
3	1	2	$\frac{41276762}{2083725}$
3	1	1	$\frac{8186375738}{31255875}$
3	1	0	$\frac{10434543841}{10418625}$
3	0	3	$\frac{115958336}{258339375}$
3	0	2	$\frac{3487886389}{172226250}$
3	0	1	$\frac{39671949511}{172226250}$
3	0	0	$\frac{315432259813}{516678750}$
2	3	0	3
2	2	2	$\frac{147968}{99225}$
2	2	1	$\frac{4065856}{99225}$
2	2	0	$\frac{112938659}{396900}$
2	1	4	$\frac{33947257}{393824025}$
2	1	3	$\frac{2164641406}{393824025}$
2	1	2	$\frac{1130666508023}{9845600625}$
2	1	1	$\frac{9041228678666}{9845600625}$
2	1	0	$\frac{5074377793279}{2187911250}$
2	0	4	$\frac{5737086433}{65101522500}$
2	0	3	$\frac{182912198807}{32550761250}$
2	0	2	$\frac{2142193804481}{21700507500}$
2	0	1	$\frac{17194597630733}{32550761250}$
2	0	0	$\frac{34399269232129}{65101522500}$
1	3	1	$\frac{544}{315}$
1	3	0	$\frac{7474}{315}$
1	2	3	$\frac{1817216}{6251175}$
1	2	2	$\frac{24359162}{2083725}$
1	2	1	$\frac{5056135954}{31255875}$
1	2	0	$\frac{47437544293}{62511750}$
1	1	5	$\frac{3399584}{393824025}$
1	1	4	$\frac{57943048}{78764805}$
1	1	3	$\frac{1934522477168}{88610405625}$
1	1	2	$\frac{16308922895827}{59073603750}$
1	1	1	$\frac{85574440190593}{59073603750}$
1	1	0	$\frac{62146993882837}{25317258750}$
1	0	5	$\frac{143632424}{16275380625}$
1	0	4	$\frac{2448093778}{3255076125}$
1	0	3	$\frac{98527787884}{5425126875}$
1	0	2	$\frac{475980699652}{3255076125}$
1	0	1	$\frac{4488365319374}{16275380625}$
1	0	0	$\frac{1086412028}{10333575}$
0	4	0	1
0	3	2	$\frac{27794}{99225}$
0	3	1	$\frac{624478}{99225}$
0	3	0	$\frac{1747307}{26460}$
0	2	4	$\frac{1744613}{78764805}$
0	2	3	$\frac{437610814}{393824025}$
0	2	2	$\frac{452220491743}{19691201250}$
0	2	1	$\frac{4035916860091}{19691201250}$
0	2	0	$\frac{10861695961357}{13127467500}$
0	1	6	$\frac{135424}{393824025}$
0	1	5	$\frac{2964608}{78764805}$
0	1	4	$\frac{962538972893}{637994920500}$
0	1	3	$\frac{43240103983367}{1594987301250}$
0	1	2	$\frac{238300444324673}{1063324867500}$
0	1	1	$\frac{170939874507539}{227855328750}$
0	1	0	$\frac{326386907513329}{65101522500}$
0	0	6	$\frac{5721664}{16275380625}$

$\deg_x$	$\deg_z$	$\deg_s$	coefficient
0	0	5	$\frac{125254688}{3255076125}$
0	0	4	$\frac{6479423564}{5425126875}$
0	0	3	$\frac{41256502232}{3255076125}$
0	0	2	$\frac{513831327844}{16275380625}$
0	0	1	$\frac{148103696}{10333575}$

22-term equilibrium-scaled homogeneous certificate ($U = A - 1/4$)

$\deg_x$	$\deg_z$	coefficient
7	0	$\frac{1}{4} + U$
6	1	$\frac{5}{4}$
6	0	$\frac{13}{2} + 22U$
5	1	$\frac{113}{4} + 3U$
5	0	$\frac{271}{4} + 183U$
4	2	$\frac{15}{4}$
4	1	$\frac{971}{4} + 44U$
4	0	$\frac{673}{2} + 714U$
3	2	$\frac{223}{4} + 3U$
3	1	$997 + 242U$
3	0	$\frac{2849}{4} + 1313U$
2	3	$\frac{15}{4}$
2	2	$311 + 22U$
2	1	$\frac{7919}{4} + 586U$
2	0	$312 + 992U$
1	3	$\frac{111}{4} + U$
1	2	$\frac{3077}{4} + 59U$
1	1	$\frac{6593}{4} + 609U$
1	0	$256U$
0	4	$\frac{5}{4}$
0	3	$\frac{299}{4}$
0	2	$\frac{3181}{4}$

84-term equilibrium-scaled spatial certificate

$\deg_x$	$\deg_z$	$\deg_s$	coefficient
7	0	0	$\frac{8281}{8100}A$
6	1	0	$\frac{8281}{24300}$
6	0	1	$\frac{160888}{91125}A$
6	0	0	$\frac{8281}{8100} + \frac{4420871}{182250}A$
5	1	1	$\frac{160888}{273375}$
5	1	0	$\frac{4420871}{546750} + \frac{8281}{2700}A$
5	0	2	$\frac{3384901}{2733750}A$
5	0	1	$\frac{160888}{91125} + \frac{96932147}{2733750}A$
5	0	0	$\frac{4420871}{182250} + \frac{1210005017}{5467500}A$
4	2	0	$\frac{8281}{8100}$
4	1	2	$\frac{3384901}{8201250}$
4	1	1	$\frac{96932147}{8201250} + \frac{321776}{91125}A$
4	1	0	$\frac{315078023}{4100625} + \frac{4420871}{91125}A$
4	0	3	$\frac{115958336}{258339375}A$
4	0	2	$\frac{3384901}{2733750} + \frac{3487886389}{172226250}A$
4	0	1	$\frac{96932147}{2733750} + \frac{45494837011}{172226250}A$
4	0	0	$\frac{1073317517}{5467500} + \frac{503404909813}{516678750}A$
3	2	1	$\frac{321776}{273375}$

$\deg_x$	$\deg_z$	$\deg_s$	coefficient
3	2	0	$\frac{4420871}{273375} + \frac{8281}{2700}A$
3	1	3	$\frac{115958336}{775018125}$
3	1	2	$\frac{3487886389}{516678750} + \frac{6251648}{4100625}A$
3	1	1	$\frac{47319306931}{516678750} + \frac{171782416}{4100625}A$
3	1	0	$\frac{578603925523}{1550036250} + \frac{2360113561}{8201250}A$
3	0	4	$\frac{5737086433}{65101522500}A$
3	0	3	$\frac{115958336}{258339375} + \frac{182912198807}{32550761250}A$
3	0	2	$\frac{3487886389}{172226250} + \frac{2513945531981}{21700507500}A$
3	0	1	$\frac{39671949511}{172226250} + \frac{29165273350733}{32550761250}A$
3	0	0	$\frac{315432259813}{516678750} + \frac{138537167092129}{65101522500}A$
2	3	0	$\frac{8281}{8100}$
2	2	2	$\frac{6251648}{12301875}$
2	2	1	$\frac{171782416}{12301875} + \frac{160888}{91125}A$
2	2	0	$\frac{4871148347}{49207500} + \frac{4420871}{182250}A$
2	1	4	$\frac{5737086433}{195304567500}$
2	1	3	$\frac{182912198807}{97652283750} + \frac{76777376}{258339375}A$
2	1	2	$\frac{2613196695629}{65101522500} + \frac{2058349189}{172226250}A$
2	1	1	$\frac{33256099805357}{97652283750} + \frac{28172026051}{172226250}A$
2	1	0	$\frac{184975683058783}{195304567500} + \frac{388034017813}{516678750}A$
2	0	5	$\frac{143632424}{16275380625}A$
2	0	4	$\frac{5737086433}{65101522500} + \frac{2448093778}{3255076125}A$
2	0	3	$\frac{182912198807}{32550761250} + \frac{119438423884}{5425126875}A$
2	0	2	$\frac{2142193804481}{21700507500} + \frac{878963880652}{3255076125}A$
2	0	1	$\frac{17194597630733}{32550761250} + \frac{21944279796374}{16275380625}A$
2	0	0	$\frac{34399269232129}{65101522500} + \frac{21842305628}{10333575}A$
1	3	1	$\frac{160888}{273375}$
1	3	0	$\frac{4420871}{546750} + \frac{8281}{8100}A$
1	2	3	$\frac{76777376}{775018125}$
1	2	2	$\frac{2058349189}{516678750} + \frac{2348593}{8201250}A$
1	2	1	$\frac{29084261011}{516678750} + \frac{52768391}{8201250}A$
1	2	0	$\frac{212816762834}{775018125} + \frac{1090212071}{16402500}A$
1	1	5	$\frac{143632424}{48826141875}$
1	1	4	$\frac{2448093778}{9765228375} + \frac{294839597}{13020304500}A$
1	1	3	$\frac{124275398572}{16275380625} + \frac{36978113783}{32550761250}A$
1	1	2	$\frac{9956722796683}{97652283750} + \frac{503148512477}{21700507500}A$
1	1	1	$\frac{11312104230233}{19530456750} + \frac{6606581721077}{32550761250}A$
1	1	0	$\frac{241757054977}{221433750} + \frac{52308962411329}{65101522500}A$
1	0	6	$\frac{5721664}{16275380625}A$
1	0	5	$\frac{143632424}{16275380625} + \frac{125254688}{3255076125}A$
1	0	4	$\frac{2448093778}{3255076125} + \frac{8242156364}{5425126875}A$
1	0	3	$\frac{98527787884}{5425126875} + \frac{86436565592}{3255076125}A$
1	0	2	$\frac{475980699652}{3255076125} + \frac{3438519744244}{16275380625}A$
1	0	1	$\frac{4488365319374}{16275380625} + \frac{7094103536}{10333575}A$
1	0	0	$\frac{1086412028}{10333575} + \frac{4999696}{6561}A$
0	4	0	$\frac{8281}{24300}$
0	3	2	$\frac{2348593}{24603750}$
0	3	1	$\frac{52768391}{24603750}$
0	3	0	$\frac{570259573}{24603750}$
0	2	4	$\frac{294839597}{39060913500}$
0	2	3	$\frac{36978113783}{97652283750}$
0	2	2	$\frac{521791643711}{65101522500}$
0	2	1	$\frac{3617447476357}{48826141875}$
0	2	0	$\frac{30203751676613}{97652283750}$
0	1	6	$\frac{5721664}{48826141875}$
0	1	5	$\frac{125254688}{9765228375}$
0	1	4	$\frac{11480941147}{21700507500}$

$\deg_x$	$\deg_z$	$\deg_s$	coefficient
0	1	3	$\frac{975299997269}{97652283750}$
0	1	2	$\frac{1716319100l769}{195304567500}$
0	1	1	$\frac{10010796373877}{32550761250}$
0	1	0	$\frac{650029271329}{65101522500}$
0	0	6	$\frac{5721664}{16275380625}$
0	0	5	$\frac{125254688}{3255076125}$
0	0	4	$\frac{6479423564}{5425126875}$
0	0	3	$\frac{41256502232}{3255076125}$
0	0	2	$\frac{513831327844}{16275380625}$
0	0	1	$\frac{148103696}{10333575}$

S8.2 Signed scalar and rational-function certificates

Each generated row below states the exact scalar polynomial or rational expression, the descending coefficient order after its indicated shift, the sign of the polynomial on its parameter domain, and whether an external negative prefactor reverses that sign. The shifts use $u = m - 3$ and $v = \nu - 1$, leaving $z = y^2$ reserved for the modulus tables. In particular, the P_C row certifies $C_m < 0$ through its external factor -215; the sign of S_m is certified separately.

Unit-profile boundary denominator. The polynomial being certified is

$$\mathcal{Q}_m = 589180301m^3 - 3500015940m^2 + 6930529579m - 4574434500.$$

After $m = u + 3$, the following coefficients are listed in descending powers of u:

$$589180301, \quad 1802606769, \quad 1838302066,$$
$$624878904$$

Every coefficient is positive, so $\mathcal{Q}_m > 0$ for $m \geq 3$. There is no external sign prefactor.

Unit-profile critical-denominator bound. The auxiliary polynomial is

$$633906000m^2 - 2491895430m + 2448652733.$$

After $m = u + 3$, the following coefficients are listed in descending powers of u:

$$633906000, \quad 1311540570, \quad 678120443$$

Every coefficient is positive, so the polynomial is positive for $m \geq 3$. There is no external sign prefactor.

Unit-profile cubic numerator lower bound. The polynomial being certified is

$$\begin{aligned} L_m = {} & 272994514782766788 6720m^5 - 27755132420474170999952m^4 \\ & + 112813395868533457497683m^3 - 229153280695458887386228m^2 \\ & + 232620996871721820873517m - 94412163900120968220300. \end{aligned}$$

This is the exact clearing polynomial defined in (S6.1). Its rational identity is

$$R_m + C_m \frac{m-2}{90m-179} = \frac{L_m}{286118780220(8m-17)(90m-179)\mathcal{Q}_m}.$$

After $m = u + 3$, the following coefficients are listed in descending powers of u:

$$2729945147827667886720, \quad 13194044796940847300848, \quad 25446870127333515303059,$$
$$24475321329207325509911, \quad 11736484608366875120374, \quad 2243933770033305838488$$

Every coefficient is positive, so $L_m > 0$ for $m \geq 3$. It enters the clearing identity with a positive denominator and no external negative prefactor.

Reference coefficient R_m. Define

$$P_R(m) = 68605040480814208768m^4 - 550882186169626030957m^3 + 1658612632937449670852m^2 - 2219226476204103501323m + 1113379274975809565700, \qquad R_m = \frac{P_R(m)}{286118780220(8m-17)\mathcal{Q}_m}.$$

After $m = u + 3$, the following coefficients of P_R are listed in descending powers of u:

$$68605040480814208768, \quad 272378299600144474259, \quad 405345143374782665711, \quad 267974666768626234894, \quad 66402795166594173768$$

Every coefficient is positive. The denominator is positive for $m \geq 3$, and there is no external negative prefactor; hence $R_m > 0$.

Unit-profile coefficient C_m. Define

$$P_C(m) = 652054120726848m^4 - 5151971981328467m^3 + 15265080924982572m^2 - 20102347725659113m + 9927281930180400,$$

$$C_m = -\frac{215P_C(m)}{11645046(8m-17)\mathcal{Q}_m}.$$

After $m = u + 3$, the following coefficients of P_C are listed in descending powers of u:

$$652054120726848, \quad 2672677467393709, \quad 4108255612276161, \quad 2806739366867294, \quad 719107361052288$$

Every coefficient is positive. Since the denominator is positive and the external prefactor is -215, this proves $C_m < 0$.

Reference cubic margin. Let $\nu = m - 2$ and define

$$P_{\mathrm{ref}}(\nu) = 3790502986637265684840\nu^5 - 974216530468600286489\nu^4 - 531035674409212188871\nu^3 - 5763861868270935 61\nu^2 + 3649732858601219\nu + 55281268032918,$$

$$D_{\mathrm{ref}}(\nu) = 715296950550(8\nu-1)(90\nu+1)(589180301\nu^3 + 35065866\nu^2 + 629431\nu + 3306).$$

The exact comparison is

$$N_m^{\mathrm{ref}} - \frac{1}{100} \geq \frac{P_{\mathrm{ref}}(\nu)}{D_{\mathrm{ref}}(\nu)}.$$

After $\nu = v + 1$, the following coefficients of P_{ref} are listed in descending powers of v:

$$3790502986637265684840, \quad 17978298402717728137711, \quad 33955060177057334483573, \quad 31899843595051464379292, \quad 14895188986348368035728, \quad 2762610207555043720056$$

Every coefficient is positive, and $D_{\mathrm{ref}}(\nu) > 0$ for $\nu \geq 1$. There is no external sign prefactor. Hence $N_m^{\mathrm{ref}} > 1/100$.

Gauge upper-bound tail. Let $\nu = m - 2$. The exact rational expression being bounded is

$$\tau_m(L) = -\frac{A_\tau}{15876(8\nu-1)B_\tau},$$

where

$$A_\tau = 1494249120\mathfrak{h}_m L\nu^2 - 69786990\mathfrak{h}_m L\nu + 108738630L\nu^2 + 1214388L\nu - 8521L - 125249670\nu^2 + 1031940\nu,$$

$$B_\tau = 32760\mathfrak{h}_m L\nu + 32760L\nu^2 + 4L - 4095\nu.$$

The signed upper-bound polynomial is

$$P_{\rm up}(\nu) = -\frac{189709065}{2}\nu^3 - 507201030\nu^2 - 935658\nu + 58481.$$

The following coefficients are listed in descending powers of ν:

$$-189709065/2, \quad -507201030, \quad -935658,$$
$$58481$$

This is a signed coefficient list, not a coefficientwise positivity certificate. Direct differentiation gives

$$P'_{\rm up}(\nu) = -\frac{569127195}{2}\nu^2 - 1014402060\nu - 935658 < 0 \qquad (\nu \geq 0),$$

and $P_{\rm up}(2) = -2789453215 < 0$. Thus $P_{\rm up}(\nu) < 0$ for $\nu \geq 2$. There is no external negative prefactor reversing this conclusion.

The signed scalar S_m. This sign is certified separately from P_C. With

$$S_m = -\frac{4(1760850\mathfrak{h}_m - 10253)}{462105}, \qquad \frac{1}{91} \leq \mathfrak{h}_m < \frac{1}{90},$$

the exact comparisons

$$\frac{1760850}{91} - 10253 > 0, \qquad \frac{4}{462105}\left(\frac{1760850}{90} - 10253\right) < \frac{1}{10},$$

give $-1/10 < S_m < 0$. The displayed expression itself carries the external negative prefactor -4, which reverses the positive bracket and proves the upper sign. Combining $N_m^{\rm ref} > 1/100$, $S_m > -1/10$, and $\tau_m(L) < 1/20$ gives $N_m(L) > 1/200$.

S9 Near-threshold control example

Return here to $a = b = 1$ and $H = I$. For scalars p, u, v, q, with u the Latin letter, define the affine-chain critical vector $r^{\rm aff} = (r_1, \ldots, r_m, r_Z)^T$ by

$$r_1 = 1, \qquad r_i = -\left(u + v\frac{m-1-i}{m-2}\right) \quad (2 \leq i \leq m-1), \qquad r_m = -p, \qquad r_Z = q.$$

The diagonal matrix $D = \mathrm{diag}(d_1, \ldots, d_m, d_Z)$ determined by $(A_m - D)r^{\rm aff} = 0$ has entries

$$d_1 = -2 + u + p + 2q,$$
$$d_2 = \frac{1 + 2p - u - v(m-3)/(m-2)}{u + v(m-3)/(m-2)}, \qquad d_i = \frac{v}{(m-2)u + v(m-1-i)} \quad (3 \leq i \leq m-1),$$
$$d_m = \frac{2u - 5p - 2q - 1}{p}, \qquad d_Z = \frac{2 - 2p - 4q}{q}.$$

For design parameters ω, θ, M, the near-threshold affine path is

$$p = \varepsilon, \quad u = 1 + (2-\omega)\varepsilon + \theta\varepsilon^2, \quad v = \omega\varepsilon - \theta\varepsilon^2,$$
$$q = \frac{1}{2} - \left(\frac{1}{2} + \omega\right)\varepsilon + \left(\theta - \frac{M}{2}\right)\varepsilon^2.$$

Then the critical vector tends to one half of the homogeneous kernel vector. The diffusion excess over the exact linear threshold is

$$d_Z - 8\sum_{j=2}^{m-1} d_j = 4\left(M + 6\omega + 3\omega^2 - \frac{\omega^2}{m-2}\right)\varepsilon^2 + O(\varepsilon^3).$$

For $(\omega, \theta, M) = (2/9, 1/2, 1)$ and $m = 3$, exact elimination yields

$$c_3(\varepsilon) = \frac{6}{1379} + \frac{421985}{11409846}\varepsilon + O(\varepsilon^2),$$

and rational remainder bounds prove $c_3(\varepsilon) > 0$ on $0 < \varepsilon \leq 10^{-3}$. This is a rigorous subcritical control example, not a universal nonlinear lower bound.

S10 Semilinear stability and robustness

On the real fixed-integrated-mass L^2 space, the positive diagonal Neumann diffusion operator has domain H^2_N, is sectorial, and generates an analytic semigroup. Its half-order fractional domain is H^1. In one space dimension, H^1 is a Banach algebra and embeds into C^0, so the quadratic mass-action Nemytskii map is smooth from H^1 to L^2. The patterned branch lies in H^2_N. The fixed-mass restriction removes the homogeneous conservation zero, and Neumann boundary conditions on an interval leave no continuous translation-neutral mode. Strict spectral negativity therefore implies local exponential asymptotic stability in H^1.

For robustness, perturb positive steady fluxes and equilibrium coordinates within the positive-equilibrium realization manifold, reconstructing the rate constants accordingly; also allow small perturbations of diffusion ratios and interval length. Use one scalar diffusion multiplier as the implicit variable. The critical eigenvalue derivative is nonzero, so the implicit function theorem continues the crossing. Low-mode gaps persist by analytic perturbation. A positive diffusion lower bound and a reaction-Jacobian upper bound give one neighborhood controlling all high modes. Smooth dependence of the reduced normal-form coefficients preserves the negative cubic sign and the branch spectral gap. This is a retuned codimension-one statement, not a claim that arbitrary nearby parameter points are already critical, and no dimension-uniform radius is claimed.

S11 Numerical protocol and independent verification

Cosine–Galerkin simulations use exact modal convolution for the quadratic reaction terms and an adaptive stiff integrator. Initial data are small perturbations of $\mathbf{1}$ in the critical direction. The first-mode amplitude is measured by the adjoint projection. Spatial and temporal refinement tables, open-format profile data, exact parameters, and all solver tolerances are included in the data archive. No numerical output is used in a proof.

The release contains independent programs for the reaction family, flux cone, all-spectrum block theorem, omission minors, principal-minor diffusion-ray theorem, diffusion criterion, contrast bounds, improved profile, Pareto family, mode certificates, harmonic corrections, cubic sign, and branch stability. Mutation tests change signs, endpoints, the factor eight, Fourier factors, and certificate coefficients and must be rejected.